

Econometrics II

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Problem Set 1

Due: February 6, 2026

Question 1. Example Code and Walkthrough

(a) - (e): Example and Walkthrough

(b) Is it surprising that the heteroskedastic and homoskedastic standard errors for the OLS are so similar?

Solution.

(c) What is the additional parameter we are estimating in part (d) using the likelihood based approach?

Solution.

(d) Show analytically why the MLE and MoM approaches yield identical parameter estimates to OLS

Solution.

(e) What computational method is "fminunc" using to find the minimum of the function (assuming the analytical gradient is not supplied)?

*Solution.***Question 2. Solo Exercise**

(a) Estimate β and its s.e. using MLE with the available data

Solution.

Parameter	Estimate
$\hat{\beta}_1$	0.519 (0.016)
$\hat{\beta}_2$	0.258 (0.013)
$\hat{\beta}_3$	0.485 (0.008)
$\hat{\beta}_4$	0.343 (0.005)

- (b) Estimate β using NLS. Can the inverse of the numerical Hessian from "fminunc" be used to construct s.e.? (Hint: 417 of Wooldridge)
- (c) Calculate s.e. for NLS using the analytical formulas in eq. 12.52 of Wooldridge (desc. below said equation).
- (d) Calculate s.e. for NLS using eq. 12.48 of Wooldridge. Calculate the score analytically, but use the Hessian provided by "fminunc". Why can't the gradient from "fminunc" be used?
- (e) Calculate s.e. for NLS using bootstrap (iter = 500). Note the command `datasample()` command can be helpful.
- (f) Why are s.e. for MLE significantly smaller than NLS?
- (g) If analytical score is too hard to construct, and bootstrap too time consuming, what can be done to construct s.e.?